

PRE-DEPLOYMENT SAFETY WORKSTATION

Stress-Testing **Clinical AI** Before it Reaches Patients

An open-source adversarial red-teaming sandbox and safety certification framework designed by a clinician, for robust clinical LLM deployments.

The Pre-Deployment **Safety Gap**

- **Static Benchmarks Fail:** Standard medical QA datasets (MedQA) evaluate recall, not interactive safety behaviors or guideline compliance.
- **High-Stakes Vulnerabilities:** Medical LLMs are prone to adversarial manipulation—such as drug-seeking behaviors and contraindication bypasses.
- **Lack of Regulatory Auditing:** FDA's SaMD guidelines mandate performance documentation, but no automated testing infrastructure exists.
- **Privacy Constraints:** Clinicians cannot deploy clinical evaluation tools that leak protected health information (PHI) to third-party clouds.

The Risk is Real

A single AI dosing error or missed emergency indicator directly impacts patient lives. Clinical models must be certified under pressure before production deployment.

Multi-Agent **Adversarial Sandbox**

Lumen establishes a non-linear simulation loop where specialized agents stress-test model behaviors.



The **Band SDK** Coordination Bus

- **Structured Task Handoffs:** Direct agent-to-agent task routing (Red-Team Adversary → Doctor AI → Patient AI → Safety Auditor).
- **Shared Clinical Context:** Shared session state (``BandSharedContext``) carries dialogue history, intercepted tool runs, and alert flags.
- **Decoupled Model Architecture:** Agents run on heterogeneous setups (cloud Gemini and local OpenVINO/Ollama medical models) communicating seamlessly.
- **Robust Local Fallbacks:** Real-time API logging with local fail-safes so auditing remains functional even in network-constrained EMR environments.

B Band SDK – Agent Communication Bus

A AI/ML API – Safety Auditor Inference

F Featherless AI – Serverless BioMistral-7B

Unique **Safety Audit** Capabilities

Lumen introduces advanced evaluation mechanics designed specifically for clinical validation.

| Disclaimer Burial

Detects if the Doctor AI buries critical safety caveats after delivering actionable, high-risk medical recommendations.

| Protocol Drift

Runs identical case scenarios multiple times to calculate safety score variance. High standard deviation flags a model as clinically unsafe.

| Standards Mapping

Intercepts actions and compiles full HL7 FHIR R4 Transaction Bundles, validating them against public HAPI FHIR servers.

Privacy-First Local Acceleration

Gateway Type	Model Used	Avg. Latency	Throughput	Safety Certification Role
Intel OpenVINO	Qwen 2.5 (INT4 CPU)	180ms	22 tok/sec	Privacy-focused on-device Doctor Agent
Featherless AI	BioMistral 7B (Serverless)	450ms	45 tok/sec	Open-source clinical reasoning Doctor
AI/ML API	Gemini 2.0 Flash	320ms	50 tok/sec	Safety Auditor & Adversarial Adversary

READY FOR CLINICAL INTEGRATION

A Pre-Deployment **Safety Gate** for Healthcare AI

Deploying safe, robust, and standards-compliant medical models in clinical workflows.

| Open-Source Foundations

Lumen leverages and credits key medical open-source libraries: the **Band SDK**, **HAPI FHIR**, **medSpacy**, **PyHealth**, and the **OMOP CDM**.

- **GitHub:** github.com/safevoice009/lumen-clinical
- **Live App:** lumen-clinical.vercel.app
- **Built with:** Antigravity IDE